The Fertility, Marriage, and Gender Equality Quandary

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December 2025

Motivation

- Three trends underpin the grand gender convergence in the last century:
  1. Falling fertility (Guinnane 2011)
  2. Declining marriage (Stevenson and Wolfers 2007)
  3. Converging gender (income) gaps (Goldin 2014)
- Many policies targeting individual outcomes
- Existing research often treat them in isolation or in pairwise combinations, lacking a unified framework

This Paper

1. Document a three-way trade-off: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically unlikely at all levels of economic development

2. Develop a unified model of marriage, fertility, and female labor supply to explain the empirical facts

3. Calibrate to the transition experience of Mexico, conduct welfare analysis, decomposition, and prediction

4. Consider a dynamic extension with endogenous gender human capital gap

Key Findings

1. Reducing women's child-rearing costs stands out as the unique policy that could mitigate the trade-off
2. Quantitative results on Mexico from 1990 to 2015:
   - → Gender-neutral TFP explains half of declining fertility and marriage
   - → Gender-biased TFP and the gender education gap reversal account for the narrowing gender income and welfare gaps
3. Gendered impacts of single parenthood result in dynamic propagation

Main contribution: new fact + unified model + quantification

Roadmap

- Empirical findings
- Model
- Quantitative analysis
- Dynamic extension
- Conclusion

Motivating Facts

Empirical Approach

1. Create panel dataset where fertility, dual parenthood, and gender income gap are consistently measured
2. Create dummy variables
   - → High fertility
   - → Widespread dual parenthood
   - → Gender income equality

   Assign values according to sample medians respectively
3. Visualize the intersections using Venn diagrams

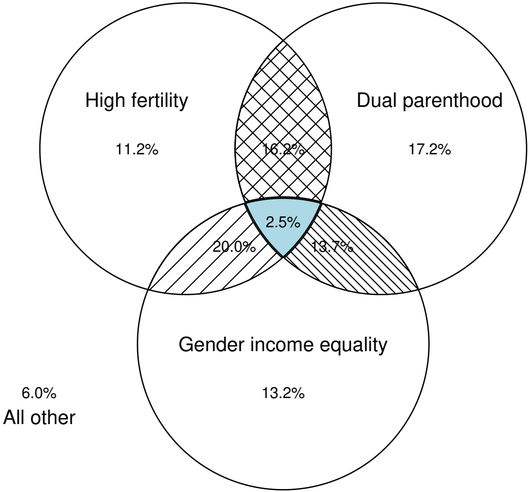
In the spirit of the dartboard approach (Ellison and Glaeser 1997), the random benchmark of the intersection is $0.5 \times 0.5 \times 0.5 = 12.5\%$

OECD Sample

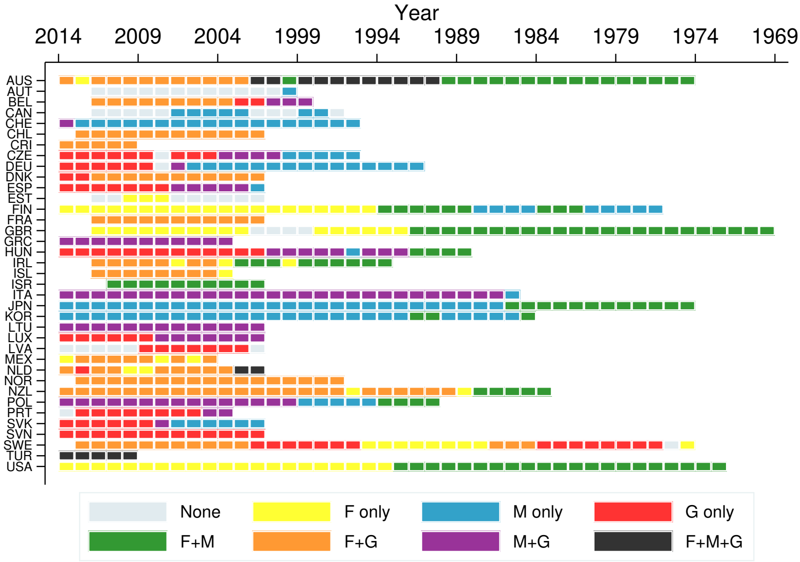
37 OECD countries from 1970 to 2014, 721 observations

- Total fertility rate (UN)
- Share of out-of-wedlock birth (OECD database)
- Gender gaps in median earnings (OECD database)

p-value=0.000 in one-sided bootstrap hypothesis test



OECD Sample



Year
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ITA
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LTU
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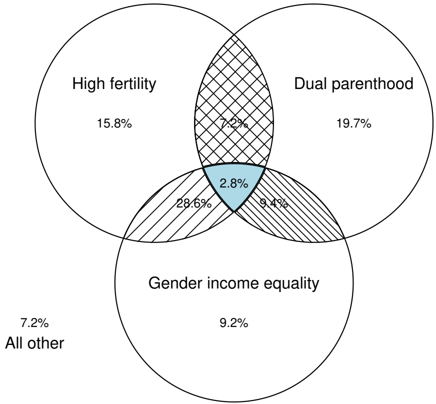
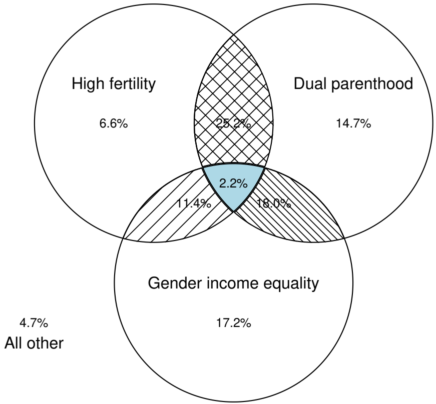
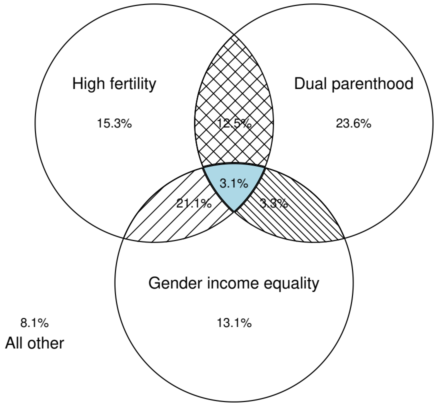
NLD
NOR
NZL
POL
PRT
SVK
SVN
SWE
TUR
USA
None
F only
M only
G only
F+M
F+G
M+G
F+M+G

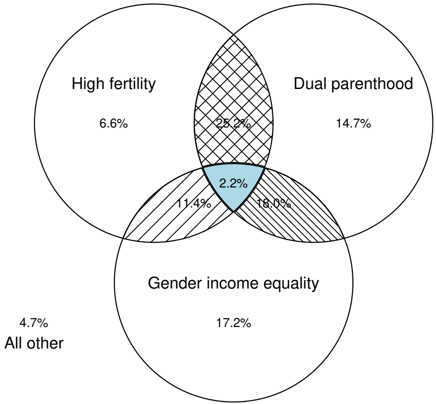
OECD Sample - High & Low Income





OECD Sample - High & Low Schooling



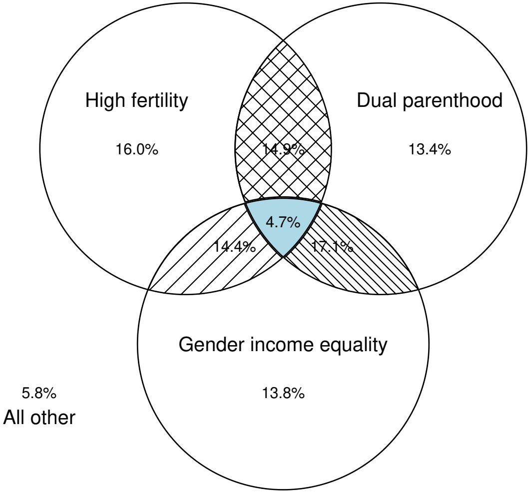


World Sample

95 countries from 1990 to 2019, 1130 observations

- Total fertility rate (UN)
- Share of children living with both parents (Brenøe and Wasserman 2025)
- Female share of labor income (World Inequality Database)

p-value=0.000 in one-sided bootstrap hypothesis test

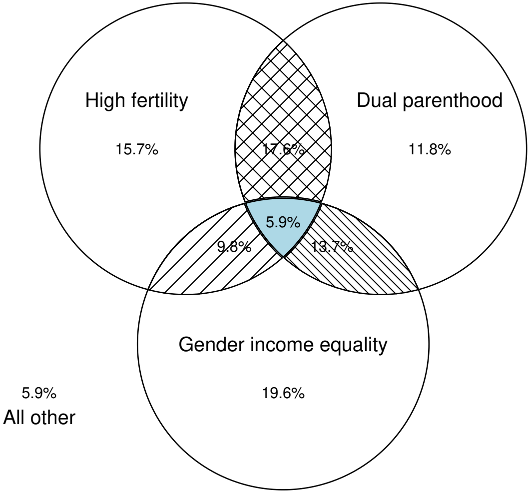


U.S. Sample

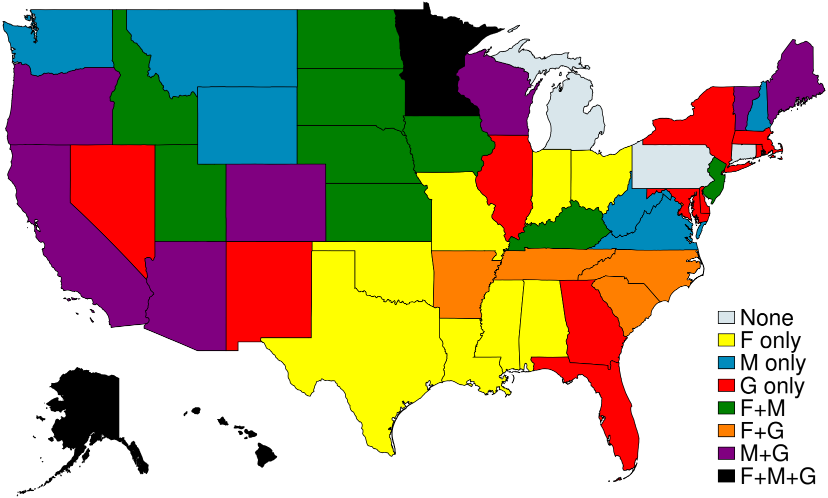
50 states and D.C. in 2023

- Live births per 1000 women aged 15-44 (CDC)
- Single-mother household share (ACS)
- Gender earnings gap (ACS)

p-value=0.029 in one-sided bootstrap hypothesis test



U.S. Sample



Questions

1. What explains the observed relationship between fertility, dual parenthood, and gender income equality?
2. Are there feasible policies that can help countries escape this trade-off?
3. What role do technological progress play in shaping the joint evolution of these three trends?

Model

Basic setup

- Becker (1973), Choo and Siow (2006)
- Individual of equal mass with gender $g \in \{\male, \female\}$ and preference

$$u(c,n) = \left((1-\beta)\cdot(c)^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}} \tag{1}$$

  where $\rho > 1$ so that $u(c,0)$ is well-defined
- Homogeneous wage within gender $w^{\male}$ and $w^{\female}$
- Wage gap is defined as

$$\Gamma^w = \frac{w^{\male}}{w^{\female}} \tag{2}$$

Marriage and Fertility – Men

- If single, men consume their labor income but have no children

$$V^{♂,s} = u(w^{♂}, 0) \tag{3}$$

- Once married, husbands work and transfer α share of income to wives

$$V^{♂,m} = u((1-\alpha)w^{♂}, n^m) \cdot \lambda \tag{4}$$

  → α is an endogenous outcome in the marriage market equilibrium
  → λ is the psychic benefit of marriage

Marriage and Fertility – Single Women

- Single female solves

$$V^{♀,s} = \max_{c^{♀,s}, l^s, n^s} \quad u(c^{♀,s}, n^s) \tag{5}$$

subject to budget and time constraints

$$c^{♀,s} = w^{♀} l^s \qquad l^s = 1 - \chi n^s$$

- Simple consumption-fertility trade-off through endogenous labor supply

Marriage and Fertility – Married Women

- Wives need to balance fertility and consumption

$$V^{♀,m} = \max_{c^{♀,m}, l^m, n^m} \quad u(c^{♀,m}, n^m) \cdot \lambda \tag{6}$$

subject to budget and time constraints

$$c^{♀,m} = \underbrace{\alpha w^{♂}}_{\text{transfer from husband}} + \underbrace{w^{♀} l^m}_{\text{own labor income}}, \qquad l^m = 1 - \chi n^m$$

- Within marriage, fertility is subject to veto $\Longrightarrow$ females determine fertility

Marriage Market Equilibrium

- Each individual draws an idiosyncratic taste shock ϵ on marriage, distributed Fréchet with scale θ
- Let $\mathcal{M}^g$ denote the share of gender g being married

$$\mathcal{M}^{\male} = \frac{1}{1 + (V^{\male,s}/V^{\male,m})^{\theta}}, \qquad \mathcal{M}^{\female} = \frac{1}{1 + (V^{\female,s}/V^{\female,m})^{\theta}}. \tag{7}$$

- Equilibrium requires

$$\mathcal{M}^{\male} = \mathcal{M}^{\female} = \mathcal{M}, \tag{8}$$

with α and n^m acting as market-clearing "prices"

Aggregate Quantities

- Aggregate fertility rate n and share of children with both parents

$$n = \mathcal{M} \cdot n^m + (1 - \mathcal{M}) \cdot n^s \tag{9}$$

$$\mathcal{D} = \frac{\mathcal{M} \cdot n^m}{n} \tag{10}$$

- Average hours worked per female is

$$l^{♀} = \mathcal{M} \cdot l^m + (1 - \mathcal{M}) \cdot l^s = 1 - \chi n \tag{11}$$

- Gender income gap

$$\Gamma^y = \frac{y^{♂}}{y^{♀}} = \frac{\Gamma^w}{l^{♀}} \tag{12}$$

Equilibrium Conditions

- A price vector (α, n^m) clears the marriage market
- For any $\mathcal{M}$, men's optimal choice yields α as a function n^m:

$$\alpha = 1 - \left[\left(\left(\frac{\mathcal{M}}{1-\mathcal{M}} \right)^{\frac{1}{\theta}} \cdot \frac{1}{\lambda} \right)^{\frac{\rho-1}{\rho}} - \frac{\beta}{1-\beta} \cdot \left(\frac{n^m}{w^{♂}} \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \tag{13}$$

  which is increasing and convex in $\alpha - n^m$ space
- The FOC for married women determines n^m as a function of α:

$$n^m \cdot \left[\left(\frac{(1-\beta)\chi}{\beta} \right)^{\rho} \cdot (w^{♀})^{\rho-1} + \chi \right] = 1 + \alpha \Gamma^w \tag{14}$$

  which is increasing and linear in $\alpha - n^m$ space

Equilibrium Characterization

- Assumption: The psychic benefit of marriage λ is sufficiently large so that the marginal man in the equilibrium does not need to be compensated if his wife decides to remain childless.
- Lemma 1: For any wage pair $\{w^{\male}, w^{\female}\}$, there exists a unique fixed point (α, n^m) that clears the marriage market
- Lemma 2: The gains from marriage are $V^{g,m}/V^{g,s} = \lambda \cdot (1 + \alpha \Gamma^w)$

Three-Way Trade-Off

- Equilibrium-implied relationship

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha \Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha \Gamma^w))^\theta}, \tag{15}$$

$$\Gamma^y = \frac{\Gamma^w}{l^{♀}}, \tag{16}$$

$$l^{♀} = 1 - \chi n. \tag{17}$$

- Main insight: three pairs of tensions:
  1. Fertility linked to gender income gap through female labor supply
  2. Fertility as part of the equilibrium price vector in the marriage market
  3. Low Γ^w improves gender income equality, but reduces gains from marriage

1. High Fertility and High Dual-parenthood

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha \Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha \Gamma^w))^\theta},$$

$$\Gamma^y = \frac{\Gamma^w}{l^{♀}},$$

$$l^{♀} = 1 - \chi n.$$

- High n implies low $l^{♀}$. To sustain high $\mathcal{M}$, Γ^w cannot be too low. Thus, Γ^y is necessarily high

2. High Fertility and Gender Income Equality

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha \Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha \Gamma^w))^\theta},$$

$$\Gamma^y = \frac{\Gamma^w}{l^{♀}},$$

$$l^{♀} = 1 - \chi n.$$

- High n again implies low $l^{♀}$. To achieve low Γ^y, Γ^w must be very small. Because α is bounded above by one, a very small wage gap reduces $\mathcal{M}$

3. High Dual-parenthood and Gender Income Equality

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha \Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha \Gamma^w))^\theta},$$

$$\Gamma^y = \frac{\Gamma^w}{l^{♀}},$$

$$l^{♀} = 1 - \chi n.$$

- Because α is bounded above, high $\mathcal{M}$ requires large Γ^w. To offset this and achieve low Γ^y, $l^{♀}$ must be high. Thus, fertility n is low

Is There Any Way Out?

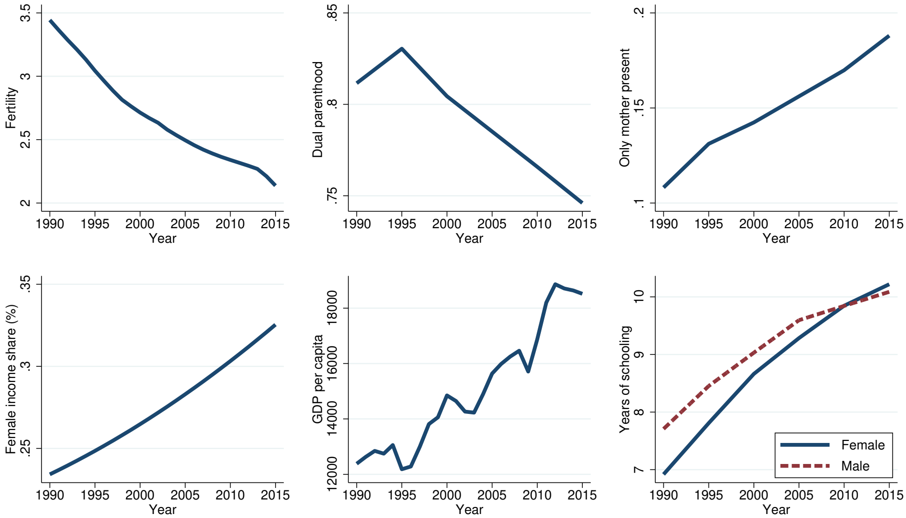
- Policy effects
  1. Pro-marriage policies (higher λ) $\Longrightarrow$ Higher $\mathcal{M}$, higher n, but also greater Γ^y
  2. Pro-equity policies (higher $w^{♀}$) $\Longrightarrow$ Achieve lower Γ^y, but lower $\mathcal{M}$ and n
  3. Pro-fertility policies (lower χ) $\Longrightarrow$ Higher $\mathcal{M}$, higher n, but reduces Γ^y because $n \cdot \chi$ falls in equilibrium
- Policies reducing child-rearing costs for women mitigate the trade-off
- One problem left: policy costs relative to GDP rise disproportionate as productivity grows

Quantitative Analysis

Data and Variable Definition

- Choose Mexico due to (1) data availability, (2) dramatic fertility decline alongside economic growth, and (3) gender education gap reversal
- Assume that $\{w_t^{\female}, w_t^{\male}\}$ as the exogenous driving force
- Decompose wage trends into three components:
  - $\rightarrow$ $A_t = w_t^{\male}$: gender-neutral productivity
  - $\rightarrow$ Γ_t^h: gender gap in human capital
  - $\rightarrow$ $B_t = \frac{w_t^{\male}/w_t^{\female}}{\Gamma_t^h}$: gender-biased productivity

Mexico's Transition



Calibration Strategy

1. Fix $\chi = 0.075$ (de La Croix and Doepke 2003)

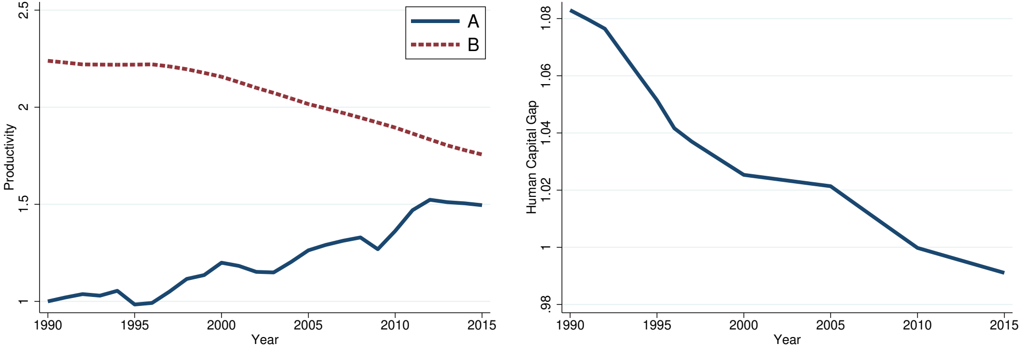
2. Compute Γ_t^h using years of schooling and Mincerian returns and back out $\{A_t, B_t\}$ from average income and female income share

3. Jointly calibrate the remaining parameters

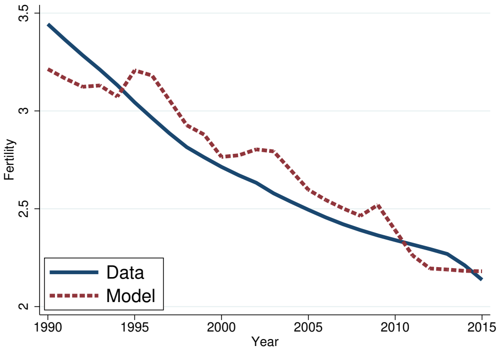
$$\underbrace{\beta = 0.104}_{n \text{ weight}}, \quad \underbrace{\rho = 1.5}_{c\text{-}n \text{ substitutability}}, \quad \underbrace{\lambda = 1.03}_{\text{psychic benefit}}, \quad \underbrace{\theta = 3.4}_{\text{shock dispersion}}$$

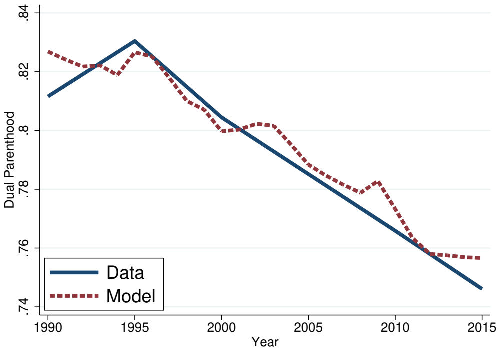
to fit trends in fertility and dual parenthood

Calibration Results (1)

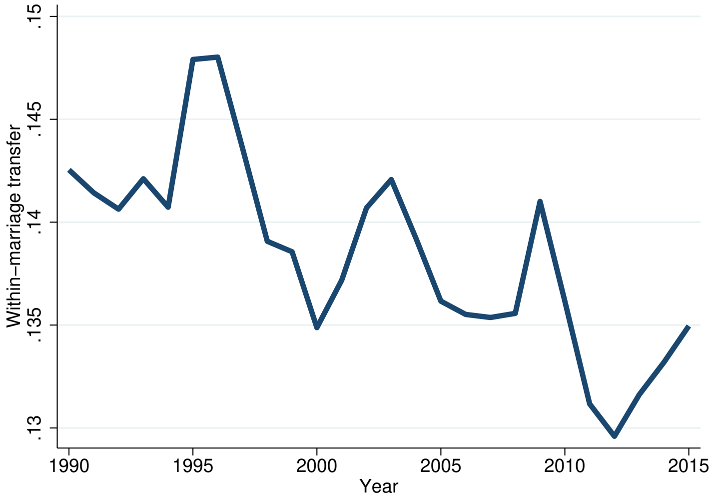


Calibration Results (2)



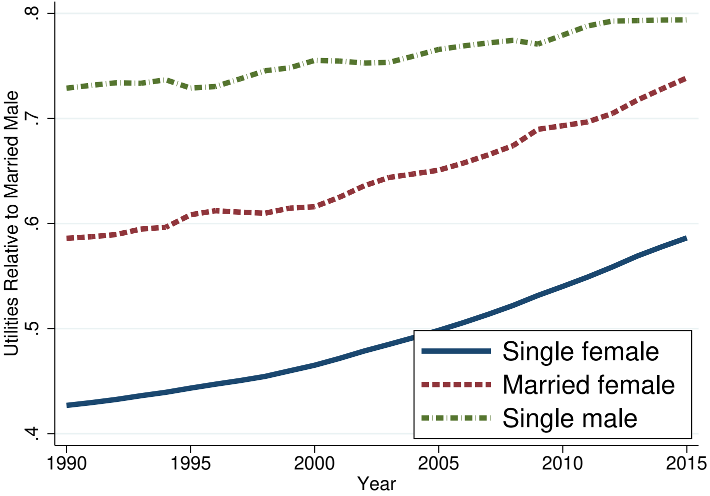


Within-Marriage Transfers

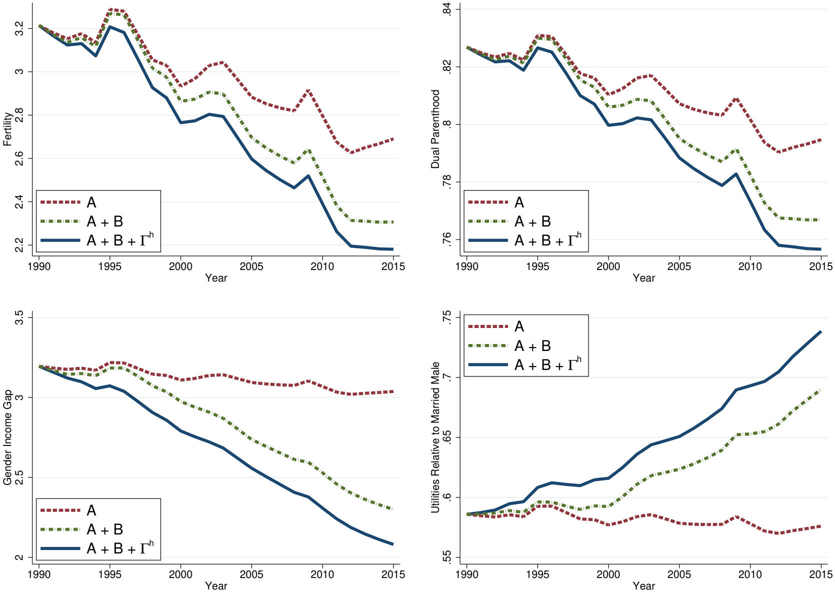


- Smaller decline in α_t relative to n_t^m because Γ_t^w falls

Relative Utilities



Decomposition



Model-Based Prediction

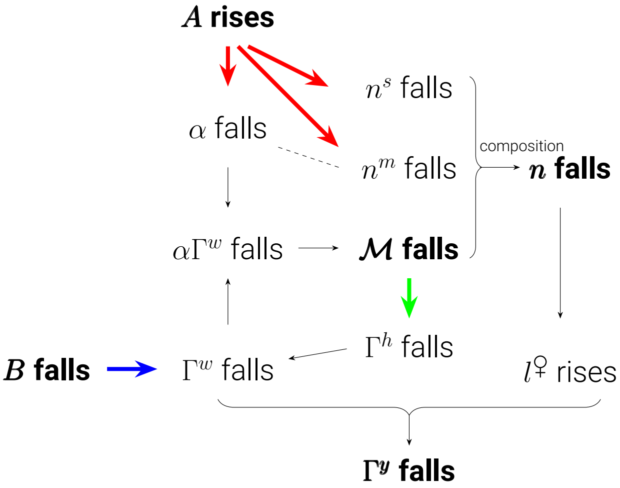
Dynamic Extension

Endogenous Gender Human Capital Gap

- Single parenthood affects the human capital of boys and girls differently
- Motivated by Bertrand and Pan (2013), Autor et al. (2019, 2023), Wasserman (2020), Reeves (2022), Frimmel et al. (2024)
- "The evidence supports an emerging consensus that growing up in a family without biological married parents produces more adverse consequences for boys than for girls." — Wasserman (2020)
- Assume that gender human capital gap depends on marriage

$$\Gamma^h_{t+1} = \mathcal{H}(\mathcal{M}_t) \qquad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0. \tag{18}$$

Dynamic Mechanism



Two exogenous forces and one propagation mechanism

Conclusion

- A three-way trade-off between fertility, marriage, and gender equality
- A unified framework to explain the empirical pattern
- Family policies offer a way out, but are getting costlier
- Quantitative analysis of Mexico's transition path
- Dynamic propagation through human capital formation